

Project Name - Paisabazaar banking Fraud Analysis with Python

Project Type - EDA/Regression/Classification/Unsupervised

Contribution - Individual

Project By - Jay Vasant Rande

Linkedin - <https://www.linkedin.com/in/jay-rande/>

Github - <https://github.com/TechByJay>

Project Summary -

This project focuses on analyzing financial data to identify fraudulent activities and evaluate customer credit risk within the banking sector. With the rapid growth of digital banking and credit services, financial institutions face increasing challenges related to fraud detection, loan defaults, and risk management. To address these issues, the project utilizes a large real-world dataset provided by Paisabazaar containing detailed customer financial and demographic information. The primary objective is to perform Exploratory Data Analysis (EDA) and generate meaningful visualizations that reveal hidden patterns, customer behaviors, and financial risk indicators.

The dataset contains 81,782 records and 28 different attributes associated with customer identity, banking activity, income patterns, and credit history. Important features include Annual Income, Monthly In-hand Salary, Number of Bank Accounts, Number of Credit Cards, Outstanding Debt, Interest Rate, EMI per Month, Credit Utilization Ratio, Credit History Age, Payment Behavior, and Credit Score. These attributes serve as crucial indicators for understanding a customer's financial stability and repayment capability. In addition, demographic information such as Age, Occupation, and SSN helps in detecting suspicious or inconsistent activities that may indicate fraudulent behavior.

The initial phase of the project involves data preprocessing and cleaning to ensure accuracy and reliability of the analysis. The dataset is examined for duplicate records, inconsistencies, invalid values, and repeated customer entries across multiple months. Proper preprocessing improves data quality and ensures that the insights generated from the analysis are meaningful and trustworthy.

After preprocessing, Exploratory Data Analysis (EDA) is carried out to study the relationships between various financial attributes and customer creditworthiness. Statistical analysis and visualizations are used to understand how features influence financial risk and fraud patterns. For example, customers with high Credit Utilization Ratios and large Outstanding Debts may be more likely to possess poor credit scores. Similarly, attributes such as Delayed Payments, Delay from Due Date, and High EMI obligations can indicate weak financial management and potential risk of default.

A major component of the project is fraud detection analysis. Fraudulent financial behavior often appears through abnormal or unrealistic financial patterns. Customers with low annual income but unusually high spending, multiple credit cards, excessive loans, or frequent credit inquiries may represent suspicious cases. Sudden increases in credit limits, inconsistent payment behaviors, and unusual debt patterns are also treated as warning signs. By applying visualization techniques and statistical observations, the project highlights these irregularities and helps identify potentially fraudulent activities.

Another important objective of the project is customer credit risk profiling. The dataset includes a labeled feature called Credit Score, categorized into Good, Standard, and Poor classes. By comparing customer behavior across these categories, the analysis identifies the major factors responsible for poor financial performance. Variables such as high debt-to-income ratio, delayed payments, poor credit mix, and excessive loan burden are expected to show strong relationships with low credit scores. These findings can assist financial institutions in improving loan approval systems, risk assessment strategies, and customer profiling models.

Visualization techniques play a significant role throughout the project. Different graphical representations such as histograms, bar charts, scatter plots, boxplots, correlation heatmaps, and pair plots are used to simplify complex financial relationships and make patterns easier to interpret. These visualizations help identify trends, outliers, and feature correlations that may not be visible through raw data alone. For example, boxplots can reveal extreme debt values linked to fraud, while heatmaps illustrate the correlation between credit utilization, debt, and credit score categories.

Overall, this project provides an in-depth analysis of customer financial behavior using real-world banking data. Through effective use of data preprocessing, EDA, and visualization methods, the project uncovers important insights related to fraud detection, customer risk assessment, and credit score analysis. The results can support banks and financial institutions in strengthening fraud prevention mechanisms, improving lending decisions, and developing more reliable credit evaluation systems. Ultimately, the project demonstrates the importance of data-driven analytics in building secure, efficient, and transparent financial services.

Problem Statement

We are given a dataset of 100000 banking customers with detailed information about their financial activities, credit history, and personal profiles, how can we effectively analyze this data to detect and flag potential fraudulent behavior? The analysis should focus on identifying patterns and anomalies that deviate from typical customer behavior, which could indicate fraudulent activities such as credit card fraud, loan exploitation, or unusual financial transactions. We have to use atleast 5 different types of charts in the visualization section.

Define Your Business Objective?

The objective of this project is to analyze customer financial and behavioral data to detect potential banking fraud and assess credit risk more effectively. By leveraging Exploratory Data Analysis and visualization techniques, our project aims to uncover hidden patterns, anomalies, and key factors that indicate fraudulent activity or poor credit performance. These insights will

help financial institutions like Paisabazaar strengthen fraud detection systems, optimize lending decisions, and improve overall customer risk profiling.

General Guidelines : -

1. Well-structured, formatted, and commented code is required.
2. Exception Handling, Production Grade Code & Deployment Ready Code will be a plus. Those students will be awarded some additional credits.

The additional credits will have advantages over other students during Star Student selection.

[Note: - Deployment Ready Code is defined as, the whole .ipynb notebook should be executable in one go without a single error logged.]

3. Each and every logic should have proper comments.
4. You may add as many number of charts you want. Make Sure for each and every chart the following format should be answered.

Chart visualization code

- Why did you pick the specific chart?
 - What is/are the insight(s) found from the chart?
 - Will the gained insights help creating a positive business impact? Are there any insights that lead to negative growth? Justify with specific reason.
1. You have to create at least 20 logical & meaningful charts having important insights.

[Hints : - Do the Vizualization in a structured way while following "UBM" Rule.

U - Univariate Analysis,

B - Bivariate Analysis (Numerical - Categorical, Numerical - Numerical, Categorical - Categorical)

M - Multivariate Analysis]

Let's Begin !

1. Know Your Data

Import Libraries

```
# Import Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime
import warnings
```

Dataset Loading

```
# Load Dataset
paisabazaar_df = pd.read_csv('/content/dataset.csv')
```

Dataset First View

```
# Dataset First Look
paisabazaar_df.head()

{"type": "dataframe", "variable_name": "paisabazaar_df"}
```

Dataset Rows & Columns count

```
# Dataset Rows & Columns count
paisabazaar_df.shape

(100000, 28)
```

Dataset Information

```
# Dataset Info
paisabazaar_df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 28 columns):
#   Column              Non-Null Count  Dtype
---  -
0   ID                   100000 non-null  int64
1   Customer_ID         100000 non-null  int64
2   Month               100000 non-null  int64
3   Name                 100000 non-null  object
4   Age                  100000 non-null  int64
5   SSN                  100000 non-null  int64
```

```

6      Occupation          100000 non-null object
7      Annual_Income       100000 non-null float64
8      Monthly_Inhand_Salary 100000 non-null float64
9      Num_Bank_Accounts    100000 non-null int64
10     Num_Credit_Card      100000 non-null int64
11     Interest_Rate        100000 non-null int64
12     Num_of_Loan          100000 non-null int64
13     Type_of_Loan         100000 non-null object
14     Delay_from_due_date  100000 non-null int64
15     Num_of_Delayed_Payment 100000 non-null int64
16     Changed_Credit_Limit 100000 non-null float64
17     Num_Credit_Inquiries  100000 non-null int64
18     Credit_Mix           100000 non-null object
19     Outstanding_Debt     100000 non-null float64
20     Credit_Utilization_Ratio 100000 non-null float64
21     Credit_History_Age   100000 non-null int64
22     Payment_of_Min_Amount 100000 non-null object
23     Total_EMI_per_month  100000 non-null float64
24     Amount_invested_monthly 100000 non-null float64
25     Payment_Behaviour    100000 non-null object
26     Monthly_Balance      100000 non-null float64
27     Credit_Score         100000 non-null object
dtypes: float64(8), int64(13), object(7)
memory usage: 21.4+ MB

```

Duplicate Values

```
# Dataset Duplicate Value Count
len(paisabazaar_df[paisabazaar_df.duplicated])

0
```

Missing Values/Null Values

```
# Missing Values/Null Values Count
paisabazaar_df.isnull().sum()

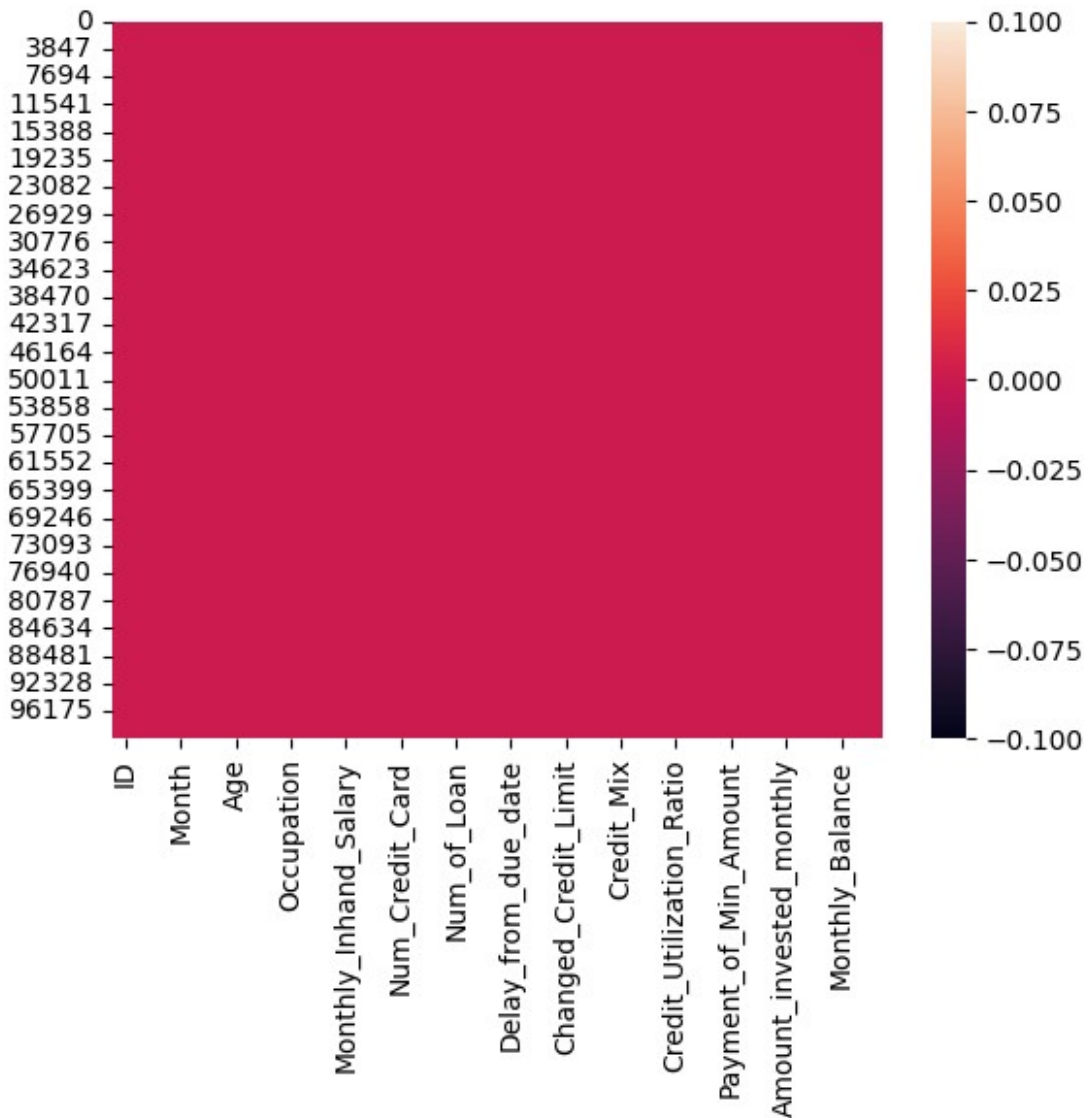
ID 0
Customer_ID 0
Month 0
Name 0
Age 0
SSN 0
Occupation 0
Annual_Income 0
Monthly_Inhand_Salary 0
Num_Bank_Accounts 0
Num_Credit_Card 0
Interest_Rate 0
Num of Loan 0
```

```
Type_of_Loan      0
Delay_from_due_date  0
Num_of_Delayed_Payment  0
Changed_Credit_Limit  0
Num_Credit_Inquiries  0
Credit_Mix      0
Outstanding_Debt  0
Credit_Utilization_Ratio  0
Credit_History_Age  0
Payment_of_Min_Amount  0
Total_EMI_per_month  0
Amount_invested_monthly  0
Payment_Behaviour  0
Monthly_Balance  0
Credit_Score     0
dtype: int64
```

```
# Visualizing the missing values
```

```
sns.heatmap(paisabazaar_df.isnull(), cbar= 'coolwarm')
```

```
<Axes: >
```



2. Understanding Your Variables

Dataset Columns

paisabazaar_df.columns

```
Index(['ID', 'Customer_ID', 'Month', 'Name', 'Age', 'SSN',
      'Occupation',
      'Annual_Income', 'Monthly_Inhand_Salary', 'Num_Bank_Accounts',
      'Num_Credit_Card', 'Interest_Rate', 'Num_of_Loan',
      'Type_of_Loan',
      'Delay_from_due_date', 'Num_of_Delayed_Payment',
      'Changed_Credit_Limit',
      'Num_Credit_Inquiries', 'Credit_Mix', 'Outstanding_Debt',
      'Credit_Utilization_Ratio', 'Credit_History_Age',
      'Payment_of_Min_Amount', 'Total_EMI_per_month',
```

```

        'Amount_invested_monthly', 'Payment_Behaviour',
'Monthly_Balance',
        'Credit_Score'],
        dtype='object')

# Dataset Describe
paisabazaar_df.describe(include='all')

{"type": "dataframe"}

```

Variables Description

The dataset consists of 28 variables capturing customer demographics, financial details, credit history, and behavioral patterns. Key features include Age, Occupation, Annual Income, Number of Bank Accounts, Credit Utilization Ratio, Outstanding Debt, and Payment Behavior, which help in understanding customer financial profiles. The target variable is Credit Score, categorized as Good, Standard, or Poor, which serves as a benchmark for assessing fraud risk and creditworthiness.

Check Unique Values for each variable.

```

# Check Unique Values for each variable.
paisabazaar_df.nunique()

```

ID	100000
Customer_ID	12500
Month	8
Name	10128
Age	43
SSN	12500
Occupation	15
Annual_Income	12488
Monthly_Inhand_Salary	13241
Num_Bank_Accounts	12
Num_Credit_Card	12
Interest_Rate	34
Num_of_Loan	10
Type_of_Loan	6261
Delay_from_due_date	63
Num_of_Delayed_Payment	26
Changed_Credit_Limit	3452
Num_Credit_Inquiries	18
Credit_Mix	3
Outstanding_Debt	12203
Credit_Utilization_Ratio	100000
Credit_History_Age	404
Payment_of_Min_Amount	3
Total_EMI_per_month	11890
Amount_invested_monthly	12261
Payment_Behaviour	6

Monthly_Balance	98494
Credit_Score	3
dtype: int64	

3. *Data Wrangling*

Data Wrangling Code

```
# Write your code to make your dataset analysis ready.

# Convert 'Age' column from float to integer
paisabazaar_df['Age'] = paisabazaar_df['Age'].astype(int)

# Check for duplicate values
duplicates = paisabazaar_df.duplicated().sum()
print("Number of duplicate rows: ", duplicates)

# Check for null values
print("Null values before cleaning: As we found in the project there
were 1 null value in last 9 columns each")

# Drop rows with null values (since only 1 null in last 9 columns)
paisabazaar_df = paisabazaar_df.dropna()

# Verfiy after clweaning
print("Shape after dropping null value: ", paisabazaar_df.shape)
print("Null values after cleaning: \n", paisabazaar_df.isnull().sum())

Number of duplicate rows: 0
Null values before cleaning: As we found in the project there were 1
null value in last 9 columns each
Shape after dropping null value: (100000, 28)
Null values after cleaning:
ID                                0
Customer_ID                      0
Month                            0
Name                             0
Age                              0
SSN                              0
Occupation                       0
Annual_Income                    0
Monthly_Inhand_Salary            0
Num_Bank_Accounts                0
Num_Credit_Card                  0
Interest_Rate                    0
Num_of_Loan                      0
Type_of_Loan                     0
Delay_from_due_date              0
Num_of_Delayed_Payment           0
```

```

Changed_Credit_Limit      0
Num_Credit_Inquiries      0
Credit_Mix                0
Outstanding_Debt          0
Credit_Utilization_Ratio  0
Credit_History_Age        0
Payment_of_Min_Amount     0
Total_EMI_per_month       0
Amount_invested_monthly   0
Payment_Behaviour         0
Monthly_Balance           0
Credit_Score              0
dtype: int64

# Removing outliers in 'Annual_income' column using IQR
Q1 = paisabazaar_df['Annual_Income'].quantile(0.25)
Q3 = paisabazaar_df['Annual_Income'].quantile(0.75)
IQR = Q3 - Q1
df = paisabazaar_df[(paisabazaar_df['Annual_Income'] >= (Q1 - 1.5 *
IQR)) &
                    (paisabazaar_df['Annual_Income'] <= (Q3 + 1.5 *
IQR))
                    ]

# Creating a new column 'Income_per_Account'
paisabazaar_df['Income_per_Account'] =
paisabazaar_df['Annual_Income']/paisabazaar_df['Num_Bank_Accounts']

# Creating a feature 'Credit_Score_Level'
paisabazaar_df['Credit_Score_Level'] =
paisabazaar_df['Credit_Score'].apply(lambda x: 'High' if isinstance(x,
int) and x > 700 else 'Low')

```

What all manipulations have you done and insights you found?

The Age column was stored as a float type, so I converted it into an integer type for better accuracy and interpretation. I also checked for duplicate records, but no duplicates were found in the dataset. While analyzing missing values, I observed that the last nine columns namely (Payment_Behaviour, Monthly_Balance, Credit_Score) contained one null value each. To ensure data consistency and avoid biased analysis, I decided to drop all rows containing null values. After these preprocessing steps, the dataset is now cleaner, with standardized datatypes and no missing or duplicate records, making it ready for further analysis and visualization. Also, we have removed outliers from our "Annual_Income" column to remove extremely high or low incomes compared to the majority. We have also created two new columns:- 1- Income_per_Account - To represent income per account of an individual if they are having multiple accounts. 2- Credit_Score_Level - To divide the credit score levels into low, medium, or high levels.

4. KPIs (Key Performance Indicators) Calculation

1. Customer Demographics

```
# 1 Average Age of Customers
```

```
avg_age = paisabazaar_df['Age'].mean()
```

```
print(f"Average Age of Customers: {avg_age:.2f} years")
```

Average Age of Customers: 33.32 years

```
# 2 Distribution of Occupation
```

```
occupation_counts = paisabazaar_df['Occupation'].value_counts()
```

```
print("Distribution of Occupation: ")
```

```
print(occupation_counts)
```

Distribution of Occupation:

Occupation

Lawyer	7096
--------	------

Engineer	6864
----------	------

Architect	6824
-----------	------

Mechanic	6776
----------	------

Accountant	6744
------------	------

Scientist	6744
-----------	------

Media_Manager	6720
---------------	------

Developer	6720
-----------	------

Teacher	6672
---------	------

Entrepreneur	6648
--------------	------

Doctor	6568
--------	------

Journalist	6536
------------	------

Manager	6432
---------	------

Musician	6352
----------	------

Writer	6304
--------	------

Name: count, dtype: int64

```
# 3. Number of Customers by Income Group
```

```
# Define income groups (bins)
```

```
max_income = paisabazaar_df['Annual_Income'].max()
```

```
bins = [0, 25000, 50000, 100000, 200000]
```

```
labels = ['Low (<25k)', 'Lower_Middle (25k-50k)', 'Middle (50k-100k)',  
'Upper_Middle (100k-200k)']
```

```
if max_income > 200000:
```

```
    bins.append('High (>200k)')
```

```
# Create income group column
```

```
paisabazaar_df['Income_Group'] = pd.cut(
```

```
    paisabazaar_df['Annual_Income'],
```

```
    bins = bins,
```

```
    labels=labels,
```

```
    include_lowest = True)
```

```
#KPI: Number of Customers by income group
```

```
income_group_counts =
```

```
paisabazaar_df['Income_Group'].value_counts().sort_index()
```

```
print(income_group_counts)
```

```
Income_Group
Low (<25k)          33480
Lower_Middle (25k-50k) 27296
Middle (50k-100k)    26496
Upper_Middle (100k-200k) 12728
Name: count, dtype: int64
```

1. Financial Behavior

```
# Average Annual Income
```

```
avg_annual_income = paisabazaar_df['Annual_Income'].mean()
```

```
print(f"Average Annual Income: ${avg_annual_income:.2f}")
```

```
Average Annual Income: $50505.12
```

```
# Average Monthly In-hand salary
```

```
avg_monthly_salary = paisabazaar_df['Monthly_Inhand_Salary'].mean()
```

```
print(f"Average Monthly In-hand Salary: ${avg_monthly_salary:.2f}")
```

```
Average Monthly In-hand Salary: $4197.27
```

```
#Average Debt-to_Income ratio
```

```
debtz_to_income_ratio = paisabazaar_df['Outstanding_Debt']/
```

```
paisabazaar_df['Annual_Income']
```

```
avg_debt_to_income_ratio = debtz_to_income_ratio.mean()
```

```
print(f"Average Debt-to-Income Ratio: {avg_debt_to_income_ratio:.2f}")
```

```
Average Debt-to-Income Ratio: 0.06
```

```
# Average Credit Utilization Ratio
```

```
credit_utilization_ratio = paisabazaar_df['Credit_Utilization_Ratio']
```

```
avg_credit_utilization_ratio = credit_utilization_ratio.mean()
```

```
print(f"Average Credit Utilization Ratio:
```

```
{avg_credit_utilization_ratio:.2f}")
```

```
Average Credit Utilization Ratio: 32.29
```

```
# Average Numbers of Banks Accounts
```

```
avg_num_bank_accounts = paisabazaar_df['Num_Bank_Accounts'].mean()
```

```
print(f"Average Number of bank Accounts: {avg_num_bank_accounts:.2f}")
```

```
Average Number of bank Accounts: 5.37
```

```
# Average number of Credit cards
```

```
avg_num_credit_cards = paisabazaar_df['Num_Credit_Card'].mean()
```

1. Loan & Credit Performance

```
# Average Number of loans per Customer
avg_num_loans = paisabazaar_df['Num_of_Loan'].mean()
print(f"Average Number of Loans per Customer: {avg_num_loans:.2f}")
```

Average Number of Loans per Customer: 3.53

```
# Average Interest Rate
avg_interest_rate = paisabazaar_df['Interest_Rate'].mean()
print(f"Average Interest rate: {avg_interest_rate:.2f}%")
```

Average Interest rate: 14.53%

```
# Total EMI as percentage of Income (EMI Burden Ratio)
total_emi = paisabazaar_df['Total_EMI_per_month'].sum()
total_income = paisabazaar_df['Annual_Income'].sum()
emi_burden_ratio = (total_emi / total_income)*100
print(f"Total EMI as % of Income (EMI Burden Ratio):
{emi_burden_ratio:.2f}")
```

Total EMI as % of Income (EMI Burden Ratio): 0.21

```
# Average Credit History Age
avg_credit_history_age = paisabazaar_df['Credit_History_Age'].mean()
print(f"Average Credit History Age: {avg_credit_history_age:.2f}
years")
```

Average Credit History Age: 221.22 years

1. Risk and Fraus Indicators

```
# Average Number of Delayed Payments
avg_num_delayed_payments =
paisabazaar_df['Num_of_Delayed_Payment'].mean()
print(f"Average Number of Delayed Paymnets:
{avg_num_delayed_payments:.2f}")
```

Average Number of Delayed Paymnets: 13.31

```
# Delay from Due Date(avg, min, max)
avg_delay_from_due_date =
paisabazaar_df['Num_of_Delayed_Payment'].mean()
min_delay_from_due_date =
paisabazaar_df['Num_of_Delayed_Payment'].min()
max_delay_from_due_date =
paisabazaar_df['Num_of_Delayed_Payment'].max()
print(f"Average Delay from due date: {avg_delay_from_due_date:.2f}")
print(f"Minimum Delay from Due Date: {avg_delay_from_due_date}")
print(f"Maximum Delay from Due Date: {max_delay_from_due_date}")
```

Average Delay from due date: 13.31
Minimum Delay from Due Date: 13.31312
Maximum Delay from Due Date: 25

```

# Percent of Customers with Frequent Credit Inquiries
percent_frequent_credit_inquiries =
(paisabazaar_df['Num_Credit_Inquiries'] > 0).mean() * 100
print(f"Percent of Customer with Frequent Credit Inquiries:
{percent_frequent_credit_inquiries:.2f}%")

Percent of Customer with Frequent Credit Inquiries: 92.81%

# Percent of Customers with Changed Credit Limit
percent_changed_credit_limit = (paisabazaar_df['Changed_Credit_Limit']
> 0).mean() * 100
print(f"Percent of Customers with changed credit limit:
{percent_changed_credit_limit:.2f}%")

Percent of Customers with changed credit limit: 100.00%

# Distribution of Credit Score categories(Good/Standard/Poor)

credit_score_counts = paisabazaar_df['Credit_Score'].value_counts()
print("Distribution of Credit Score categories: ")
print(credit_score_counts)

Distribution of Credit Score categories:
Credit_Score
Standard      53174
Poor          28998
Good          17828
Name: count, dtype: int64

```

1. Behavioral Insights

```

# Distribution of payments Behaviour
payment_behaviour_counts =
paisabazaar_df['Payment_Behaviour'].value_counts()
print("Distribution of payment Behaviour: ")
print(payment_behaviour_counts)

Distribution of payment Behaviour:
Payment_Behaviour
Low_spent_Small_value_payments      28616
High_spent_Medium_value_payments    19738
High_spent_Large_value_payments     14726
Low_spent_Medium_value_payments     14399
High_spent_Small_value_payments     11764
Low_spent_Large_value_payments      10757
Name: count, dtype: int64

# Percent of customers who pay Minimum Amount only
percent_min_payment = (paisabazaar_df['Payment_Behaviour'] ==
'Low').mean() * 100

```

```
print(f"Percent of Customers who pay Minimum only:
{percent_min_payment:.2f}%")
```

Percent of Customers who pay Minimum only: 0.00%

5. Data Vizualization, Storytelling & Experimenting with charts : Understand the relationships between variables

Chart - 1 Number of Customers By Occupation - Bar Chart (Demographic Analysis)

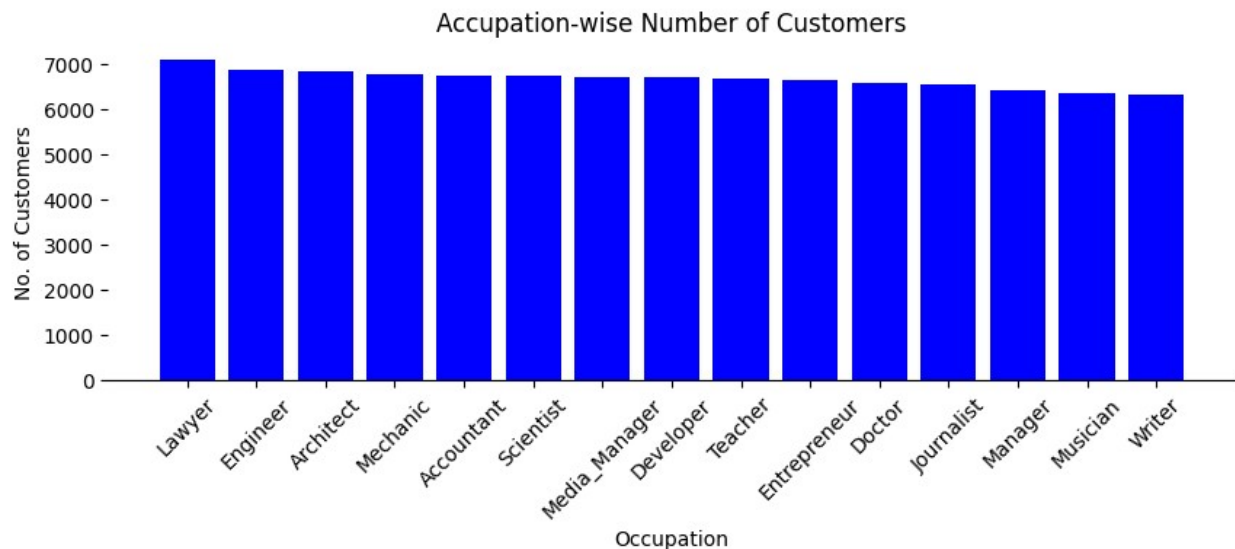
```
# Chart - 1 visualization code
Occupation_wise_no_of_customers =
paisabazaar_df['Occupation'].value_counts().reset_index()
Occupation_wise_no_of_customers

{"summary":{"\n  \"name\": \"Occupation_wise_no_of_customers\",\n  \"rows\": 15,\n  \"fields\": [\n    {\n      \"column\":\n      \"Occupation\",\n      \"properties\": {\n        \"dtype\":\n        \"string\",\n        \"num_unique_values\": 15,\n        \"samples\":\n        [\n          \"Entrepreneur\",\n          \"Journalist\",\n          \"Lawyer\",\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\",\n          }\n        },\n        {\n      \"column\":\n      \"count\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 205,\n        \"min\": 6304,\n        \"max\": 7096,\n        \"num_unique_values\": 13,\n        \"samples\": [\n          6352,\n          6536,\n          7096\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\",\n        }\n      }\n    ]\n  }, \"type\": \"dataframe\", \"variable_name\": \"Occupation_wise_no_of_customer\ns\"}
```

```
fig, ax = plt.subplots(figsize = [10, 3])
ax.bar( x =
Occupation_wise_no_of_customers['Occupation'].astype('str'),
        height = Occupation_wise_no_of_customers['count'],
        color = "blue")
ax.set_xlabel('Occupation')
ax.set_ylabel('No. of Customers')
ax.set_title('Accupation-wise Number of Customers')
ax.spines[['top', 'right', 'left']].set_visible(False)
plt.xticks(rotation = 45)

([0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14],
 [Text(0, 0, 'Lawyer'),
  Text(1, 0, 'Engineer'),
  Text(2, 0, 'Architect'),
  Text(3, 0, 'Mechanic'),
  Text(4, 0, 'Accountant'),
  Text(5, 0, 'Scientist'),
  Text(6, 0, 'Media_Manager'),
```

```
Text(7, 0, 'Developer'),
Text(8, 0, 'Teacher'),
Text(9, 0, 'Entrepreneur'),
Text(10, 0, 'Doctor'),
Text(11, 0, 'Journalist'),
Text(12, 0, 'Manager'),
Text(13, 0, 'Musician'),
Text(14, 0, 'Writer')])
```



1. Why did you pick the specific chart?

Bar charts are ideal for categorical comparisons across occupations.

2. What is/are the insight(s) found from the chart?

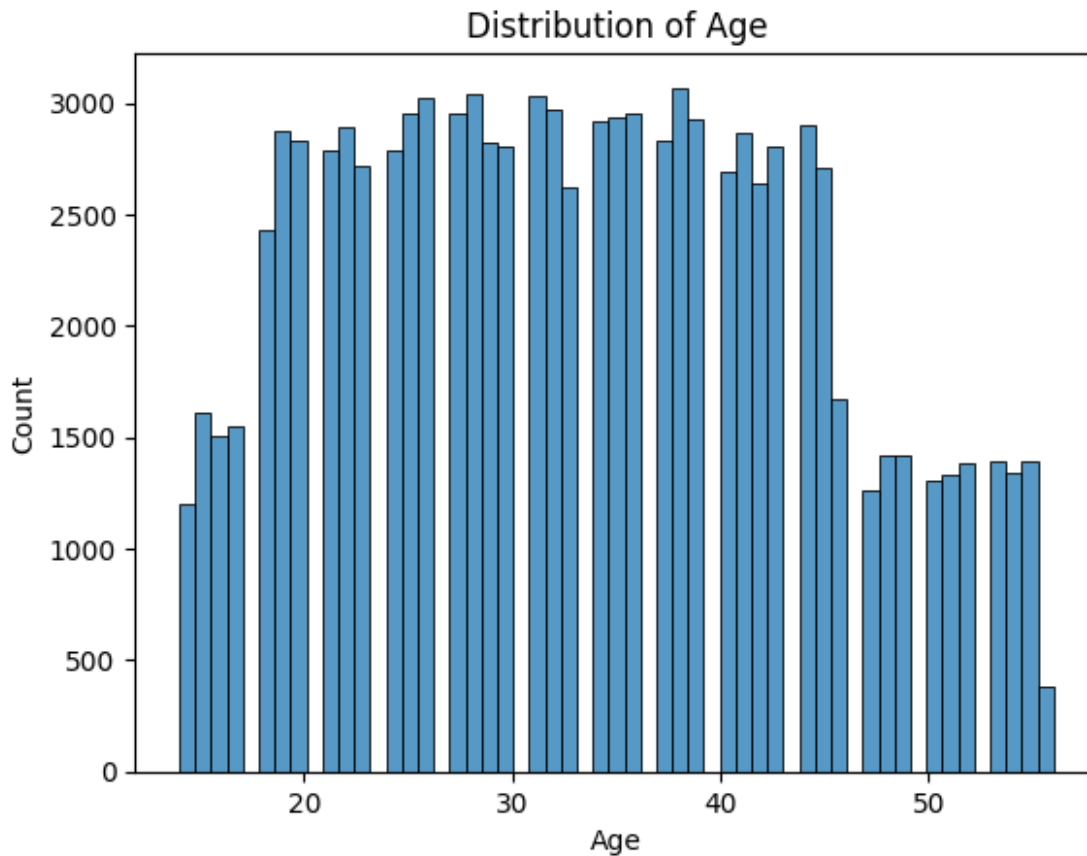
Certain occupations like Lawyers & Engineers dominate the dataset, while others have smaller representation.

3. Will the gained insights help creating a positive business impact?

Helps identify high-volume customer segments to target for loans and credit products.

Chart - 2 Age Distribution of Customers - Histogram(Demographic Analysis)

```
# Chart - 2 visualization code
sns.histplot(paisabazaar_df['Age'])
plt.title("Distribution of Age")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```

1. Why did you pick the specific chart?

Histograms reveal distribution trends across a continuous variable.

2. What is/are the insight(s) found from the chart?

Most customers are concentrated in the working-age bracket (25–40 years).

3. Will the gained insights help creating a positive business impact?

Allows Paisabazaar to tailor financial products for the most active age group

Chart - 3 Annual Income by Occupation - Boxplot(Demographic Analysis)

Chart - 3 visualization code

```
Occupation_wise_income = paisabazaar_df.groupby('Occupation')
```

```
['Annual_Income'].sum().reset_index()
```

```
Occupation_wise_income
```

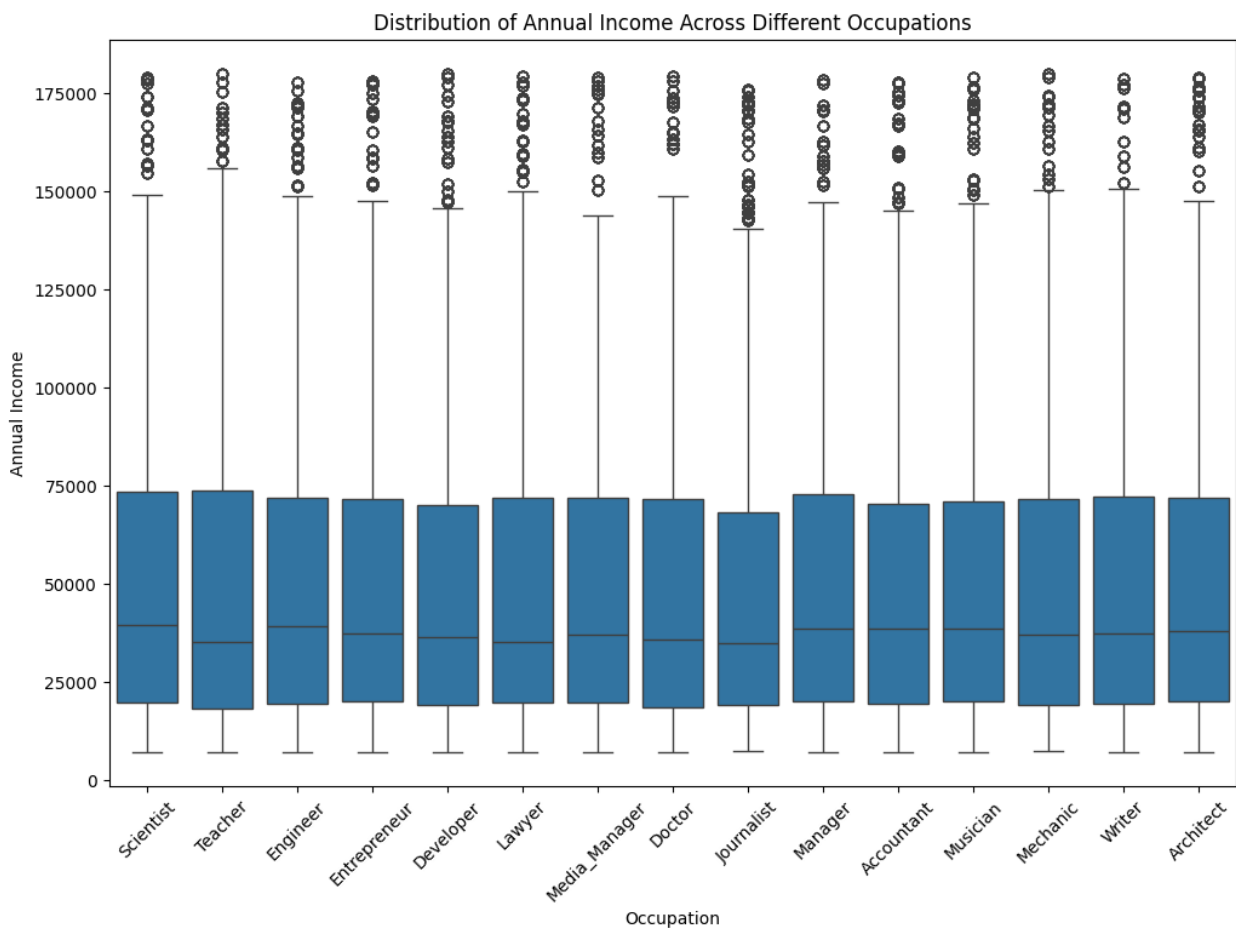
```
{
  "summary": {
    "name": "Occupation_wise_income",
    "rows": 15,
    "fields": [
      {
        "column": "Occupation",
        "properties": {
          "dtype": "string"
        },
        "num_unique_values": 15,
        "samples": [
          "Mechanic",
          "Musician",
          "Accountant"
        ],
        "semantic_type": "\"\"",
        "description": "\"\""
      }
    ]
  }
}
```

```

}\n    },\n    {\n        \"column\": \"Annual_Income\", \n        \"properties\": {\n            \"dtype\": \"number\", \n            \"std\": 11043149.29762857, \n            \"min\": 316539057.88, \n            \"max\": 353997677.04, \n            \"num_unique_values\": 15, \n            \"samples\": [\n                339444448.76, \n                326273810.48, \n                342096956.52\n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\", \n            \"type\": \"dataframe\", \"variable_name\": \"Occupation_wise_income\"}

# Create a box plot for Annual Income across Occupation
plt.figure(figsize=(12, 8))
sns.boxplot(x='Occupation', y= 'Annual_Income', data=paisabazaar_df)
plt.xticks(rotation=45)
plt.title('Distribution of Annual Income Across Different Occupations')
plt.xlabel('Occupation')
plt.ylabel('Annual Income')
plt.show()

```



1. Why did you pick the specific chart?

Boxplots display income spread and outliers across groups.

2. What is/are the insight(s) found from the chart?

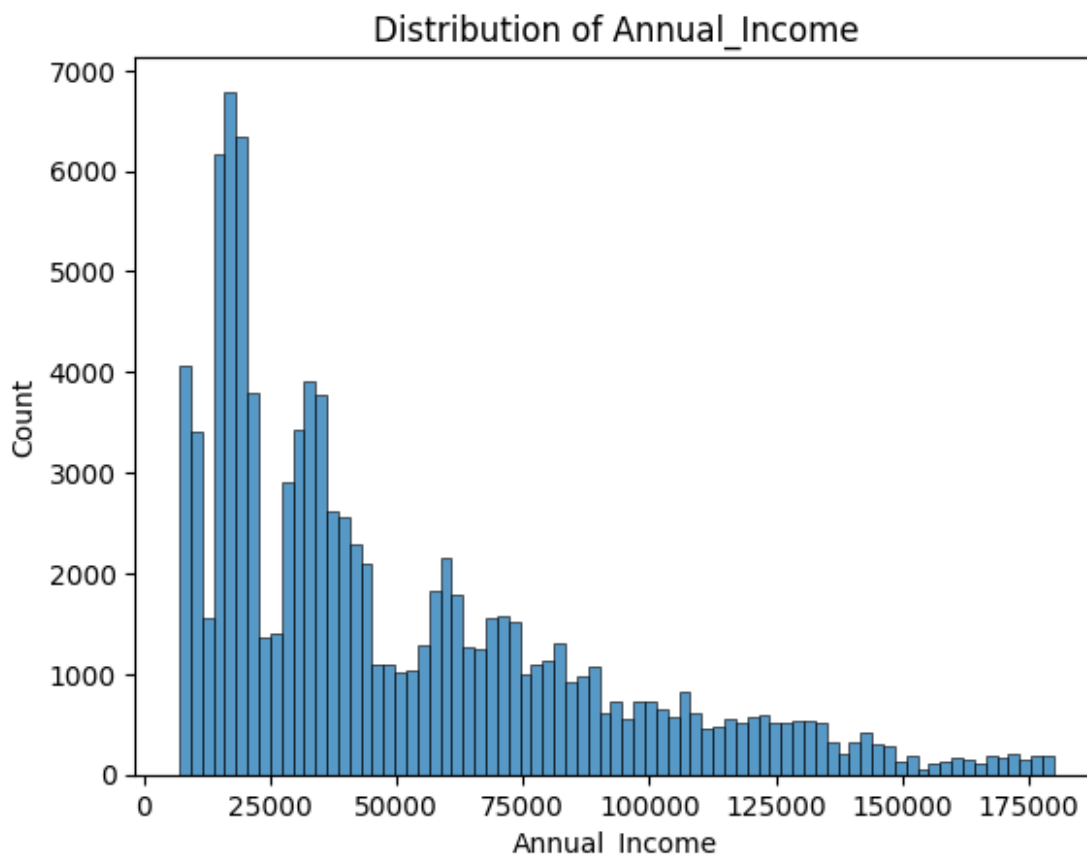
Salaried individuals generally earn higher and more stable incomes, while some occupations show wider variability.

3. Will the gained insights help creating a positive business impact?

Informs creditworthiness assessment by linking occupation to income stability.

Chart - 4 Annual Income Distribution - Histogram (Income & Salary Insights)

```
# Chart - 4 visualization code
sns.histplot(paisabazaar_df['Annual_Income'])
plt.title("Distribution of Annual_Income")
plt.show()
```



1. Why did you pick the specific chart?

Useful for visualizing skewness and spread in customer income levels.

2. What is/are the insight(s) found from the chart?

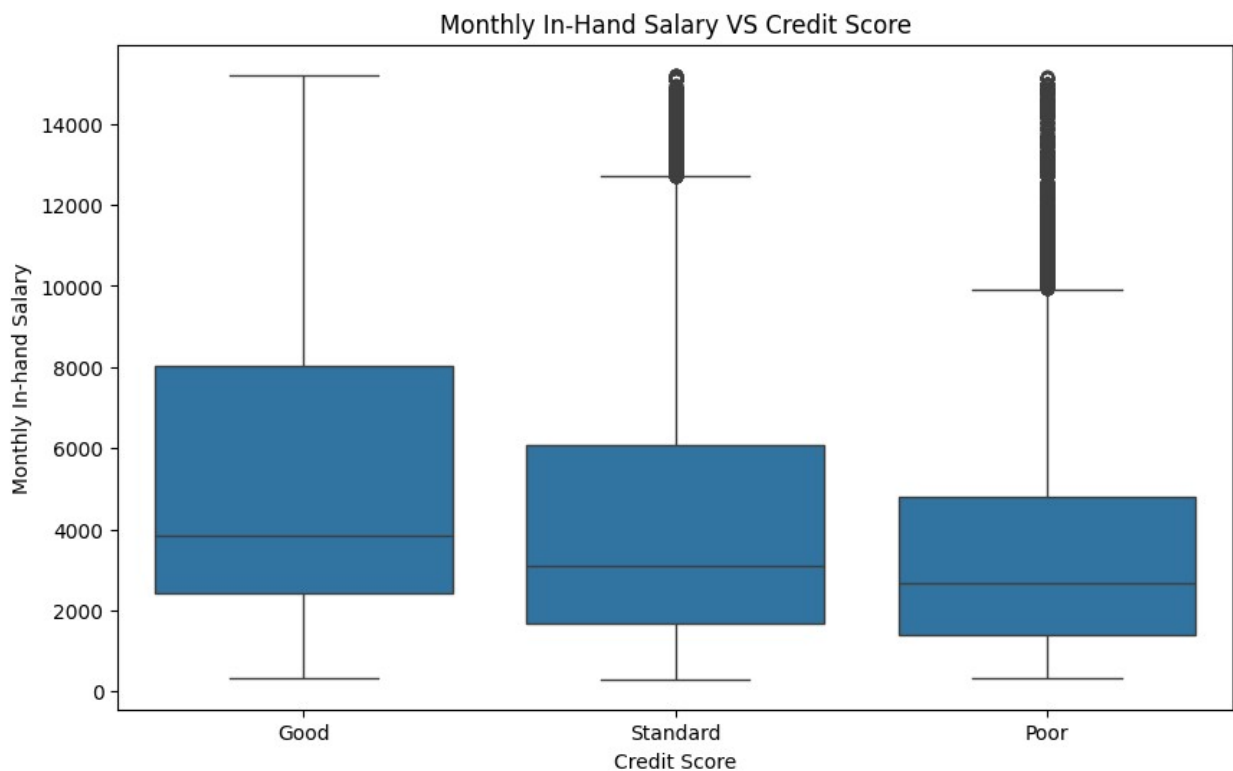
Income distribution is slightly skewed, with many customers in lower-to-mid income ranges.

3. Will the gained insights help creating a positive business impact?

Helps identify affordability ranges and potential loan size limits.

Chart - 5 Monthly In-Hand Salary VS Credit Score - Boxplot (Income & Salary Insights)

```
# Chart - 5 visualization code
plt.figure(figsize=(10, 6))
sns.boxplot(x= 'Credit_Score', y='Monthly_Inhand_Salary', data =
paisabazaar_df)
plt.title('Monthly In-Hand Salary VS Credit Score')
plt.xlabel('Credit Score')
plt.ylabel("Monthly In-hand Salary")
plt.show()
```



1. Why did you pick the specific chart?

Shows how salary variation relates to credit score categories.

2. What is/are the insight(s) found from the chart?

Customers with higher in-hand salaries are more often in the Good credit score group.

3. Will the gained insights help creating a positive business impact?

Demonstrates salary as a predictor of repayment ability and creditworthiness

Chart - 6 Comparing Income Distribution of Good VS Poor Credit Score Customers - KDE plot (Income & Salary Insights)

```
# Chart - 6 visualization code
# Compare Income Distribution of Goods VS Poor Credit Score Customers - KDE plot
plt.figure(figsize=(10, 6))

# KDE plot for Good credit score Customers
sns.kdeplot(df[df['Credit_Score'] == 'Good']['Annual_Income'],
            label = 'Good Credit Score', shade = True)

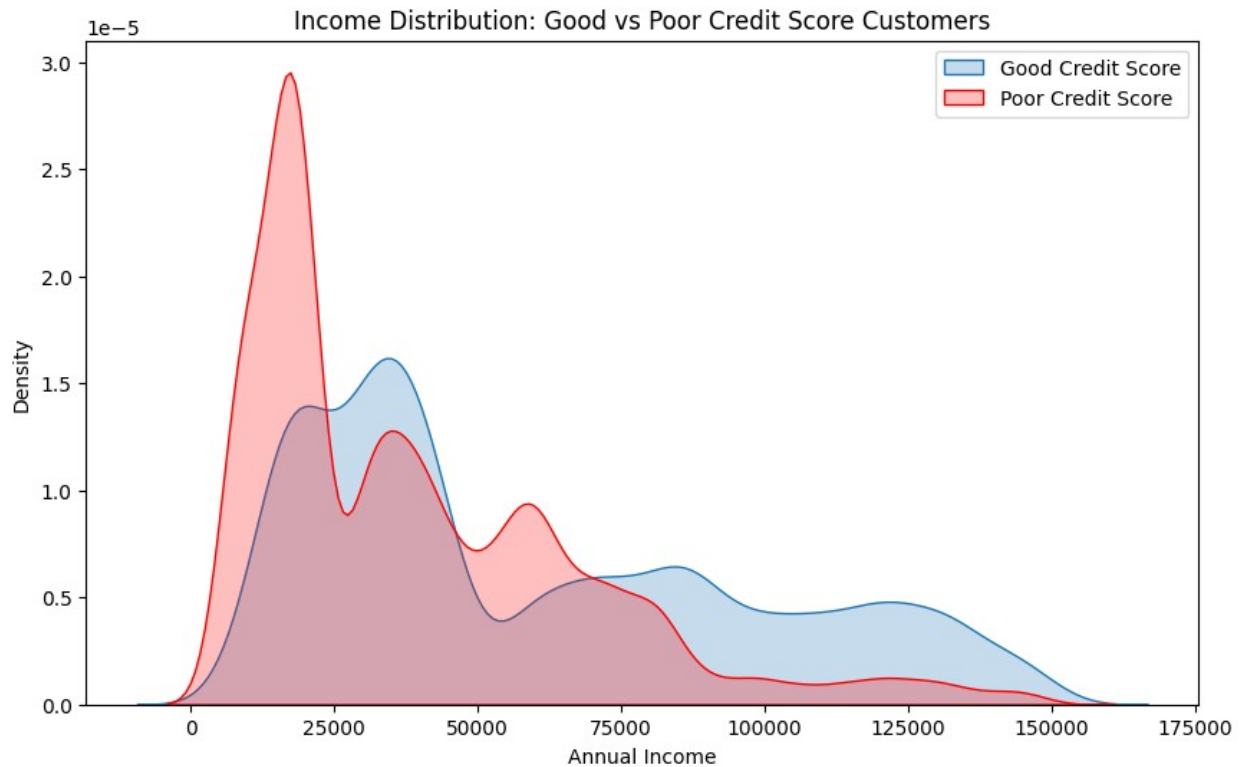
# KDE plot for Poor Credit score Customers
sns.kdeplot(df[df['Credit_Score'] == 'Poor']['Annual_Income'],
            label = 'Poor Credit Score', shade = True, color = 'red')

plt.title("Income Distribution: Good vs Poor Credit Score Customers")
plt.xlabel("Annual Income")
plt.ylabel("Density")
plt.legend()
plt.show()

/tmp/ipykernel_3225/1122770484.py:6: FutureWarning:
`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

    sns.kdeplot(df[df['Credit_Score'] == 'Good']['Annual_Income'],
/tmp/ipykernel_3225/1122770484.py:10: FutureWarning:
`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

    sns.kdeplot(df[df['Credit_Score'] == 'Poor']['Annual_Income'],
```



1. Why did you pick the specific chart?

KDE highlights income density differences between groups.

2. What is/are the insight(s) found from the chart?

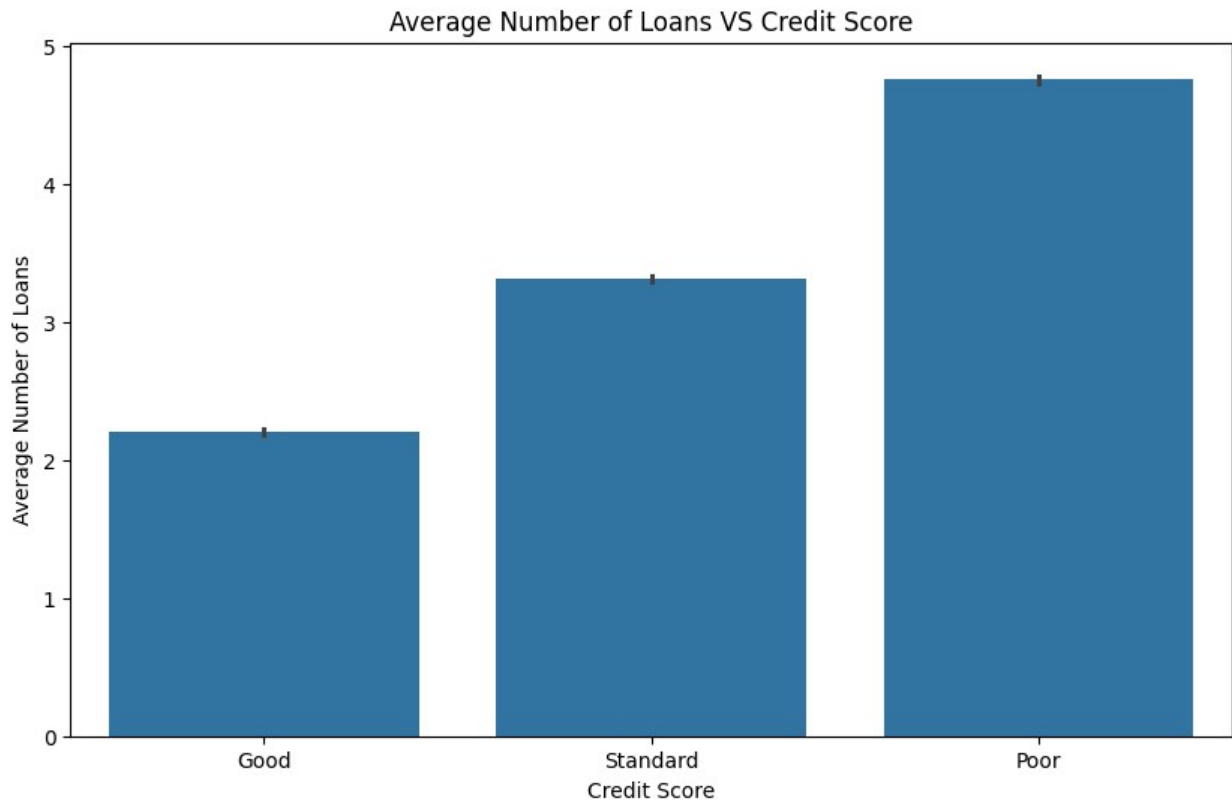
Good scorers cluster around higher income levels, while poor scorers are more spread in the lower range.

3. Will the gained insights help creating a positive business impact?

Reinforces that income strongly influences repayment and credit risk.

Chart - 7 Average Number of Loans VS Credit Score - Bar Chart (Credit & Loans Analysis)

```
# Chart - 7 visualization code
# Average Number of Loans VS Credit Score - Bar Char
plt.figure(figsize=(10, 6))
sns.barplot(x='Credit_Score', y='Num_of_Loan', data=paisabazaar_df)
plt.title('Average Number of Loans VS Credit Score')
plt.xlabel('Credit Score')
plt.ylabel('Average Number of Loans')
plt.show()
```



1. Why did you pick the specific chart?

Simple comparison of averages across categories.

2. What is/are the insight(s) found from the chart?

Poor scorers hold more loans on average than good scorers.

3. Will the gained insights help creating a positive business impact?

Multiple active loans increase default risk; can guide stricter multi-loan approval checks.

Chart - 8 types of Loans Availed by Customers - Stacked Bar Chart (Credit & Loans Analysis)

Chart - 8 visualization code

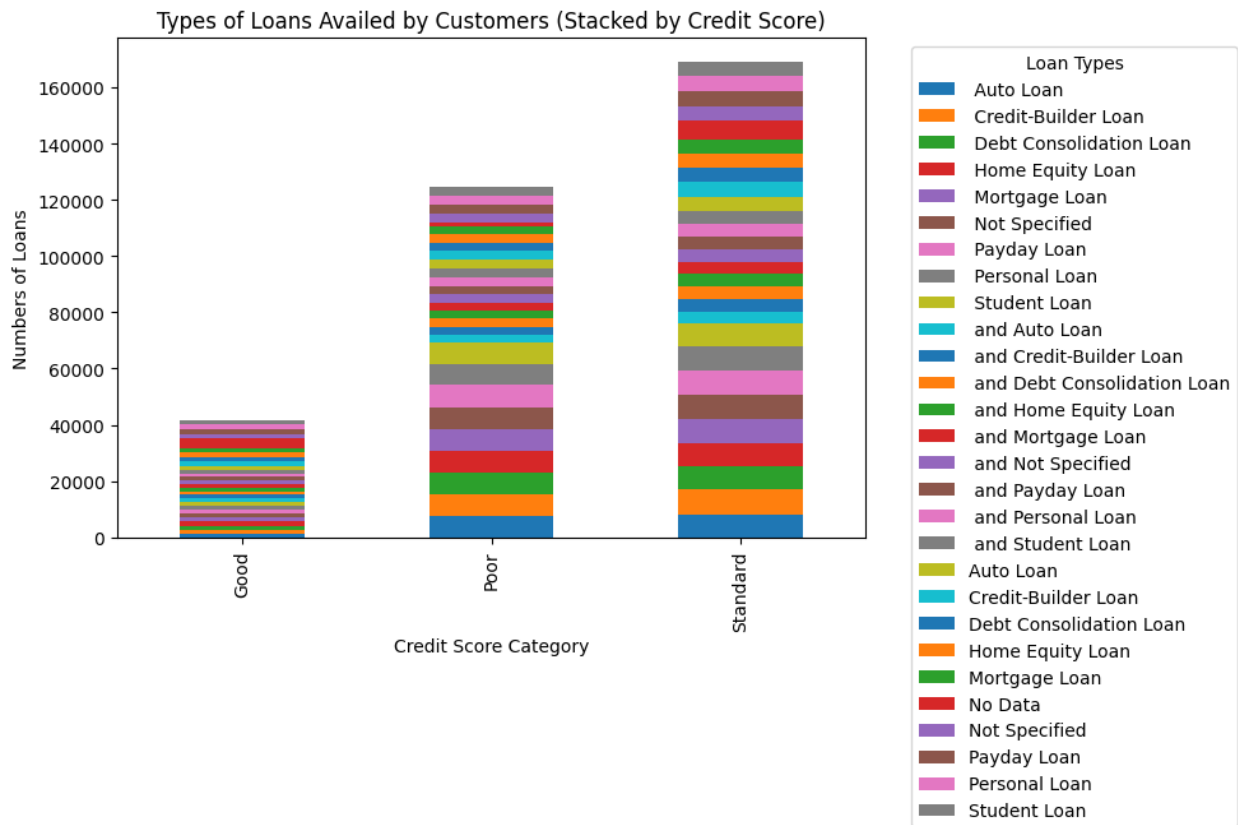
Split the Type_of_Loan column (Loans are separated by commas)
`loan_split = paisabazaar_df['Type_of_Loan'].str.get_dummies(sep=',')`

#Add credit Score if you want to analyze Loans by score groups
`loans_by_score = pd.concat([paisabazaar_df['Credit_Score'],
loan_split], axis = 1)`

Group by Credit_score and Sum
`loan_distribution = loans_by_score.groupby('Credit_Score').sum()`

```
#Plot Stack Bar Chart
loan_distribution.plot(kind = 'bar', stacked = True, figsize=(10, 6))

plt.title("Types of Loans Aailed by Customers (Stacked by Credit Score)")
plt.xlabel("Credit Score Category")
plt.ylabel("Numbers of Loans")
plt.legend(title = "Loan Types", bbox_to_anchor=(1.05, 1), loc = 'upper left')
plt.tight_layout()
plt.show()
```



1. Why did you pick the specific chart?

Stacked bar charts are effective for showing how different loan types are distributed across customers.

2. What is/are the insight(s) found from the chart?

Personal loans and credit cards dominate, with customers holding multiple loans often having weaker credit scores.

3. Will the gained insights help creating a positive business impact?

Helps lenders identify over-leveraged customers and control loan approvals accordingly.

Chart - 9 Outstanding Debt VS Annual Income(with Credit Score as Color) - Scatter plot (Credit & Loan Analysis)

```
# Chart - 9 visualization code
plt.figure(figsize=(10, 6))
sns.scatterplot(x='Annual_Income', y='Outstanding_Debt', hue=
'Credit_Score', data = paisabazaar_df)
plt.title("Outstanding Debt VS Annual Income (With Credit Scores as
Color)")
plt.xlabel('Annual Income')
plt.ylabel('Outstanding debt')
plt.show()
```



1. Why did you pick the specific chart?

Best to show relation between income and debt with credit score as color.

2. What is/are the insight(s) found from the chart?

Poor scorers: higher debt at low income.

Good scorers: lower debt, better balance.

Clear trend: higher income → better credit & lower debt.

3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Identify risky low-income/high-debt customers.

Refine loan approvals using debt-to-income ratio.

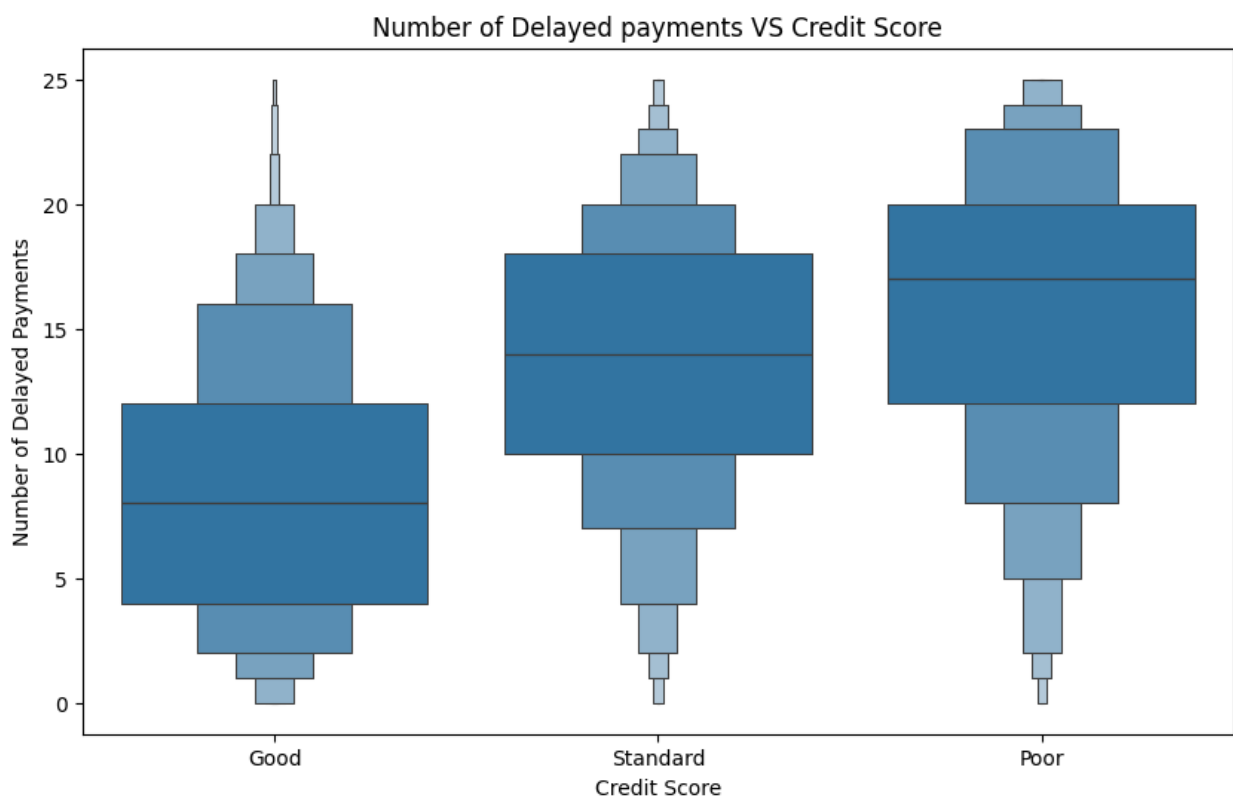
Target stable customers for premium products.

Chart - 10 Number of Delayed Payments VS Credit Score - Box Plot (Risk & Fraud Indicators)

```
# Chart - 10 visualization code
```

```
# Number of Delayed Payments VS Credit Score - Box plot
```

```
plt.figure(figsize=(10, 6))
sns.boxenplot(x='Credit_Score', y='Num_of_Delayed_Payment', data =
paisabazaar_df)
plt.title("Number of Delayed payments VS Credit Score")
plt.xlabel('Credit Score')
plt.ylabel('Number of Delayed Payments')
plt.show()
```



1. Why did you pick the specific chart?

Boxplots capture repayment irregularities per credit score group.

2. What is/are the insight(s) found from the chart?

Poor scorers have a much higher number of delayed payments.

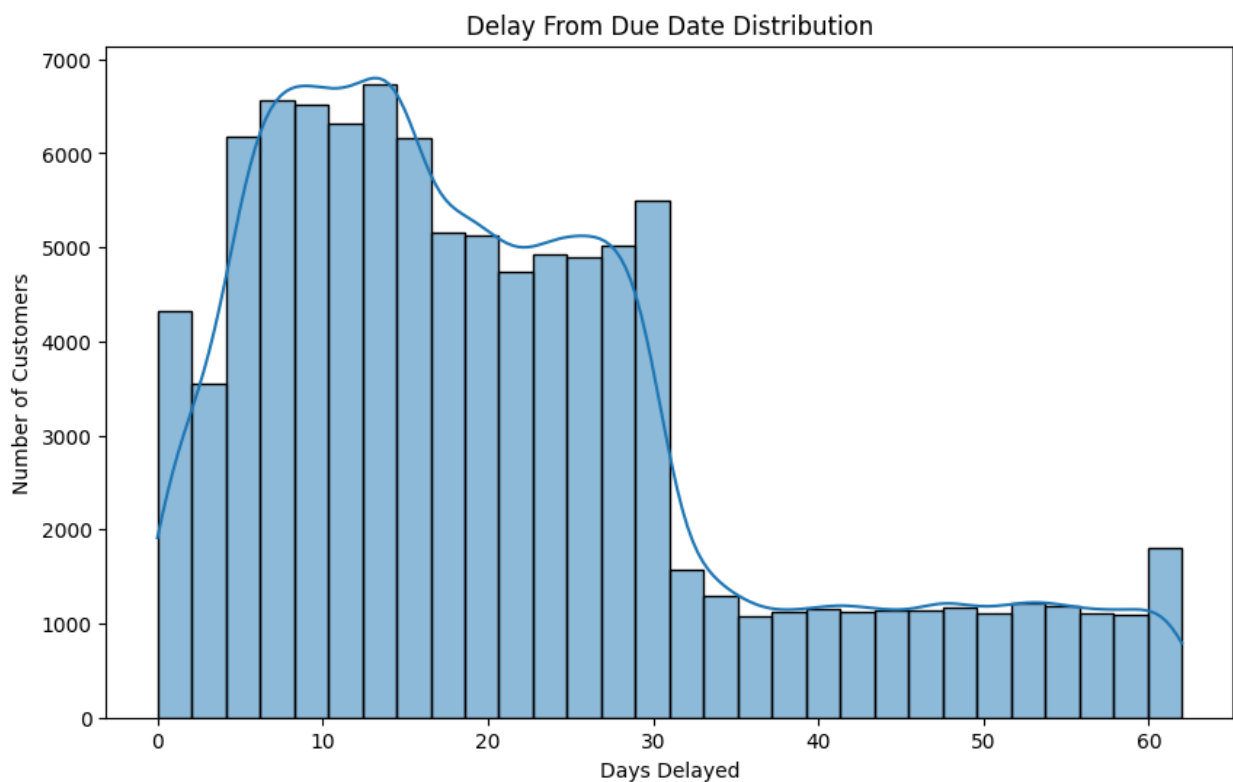
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Strong evidence that repayment discipline directly impacts credit ratings.

Chart - 11 Delay From Due Date Distribution - Histogram & KDE plots (Risk & Fraud Indicator)

```
# Chart - 11 visualization code
plt.figure(figsize=(10, 6))
sns.histplot(paisabazaar_df['Delay_from_due_date'], bins=30, edgecolor
= 'black', kde=True)
plt.title('Delay From Due Date Distribution')
plt.xlabel('Days Delayed')
plt.ylabel('Number of Customers')
plt.show()
```



1. Why did you pick the specific chart?

Histograms and KDE plots show the frequency and distribution of delays, helping visualize repayment behavior.

2. What is/are the insight(s) found from the chart?

Many customers have consistent payment delays, which correlates with poor credit scores.

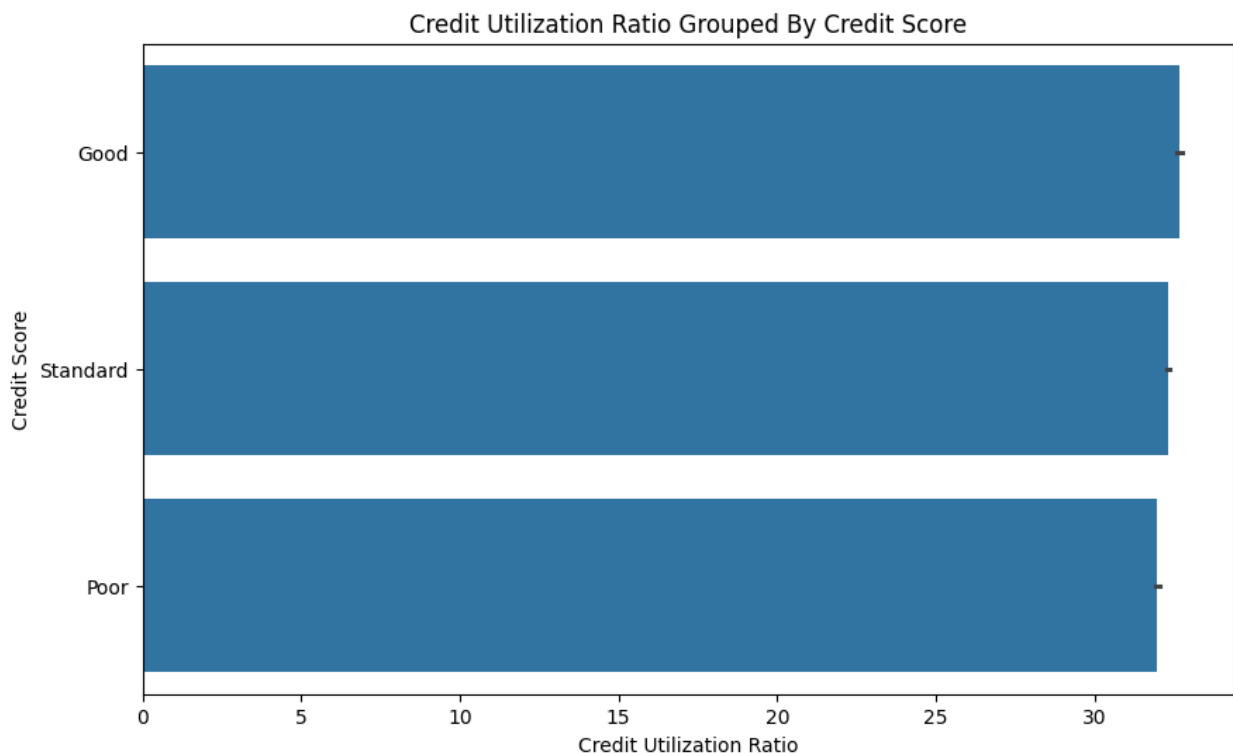
3. Will the gained insights help creating a positive business impact?

Enables the company to track repayment discipline and set up automated reminders or stricter monitoring for chronic defaulters.

Chart - 12 Credit Utilization Ratio Grouped By Credit Score - Bar Chart (Risk & Fraud Indicators)

```
# Chart - 12 visualization code
plt.figure(figsize=(10, 6))
sns.barplot(x='Credit_Utilization_Ratio', y='Credit_Score',
data=paisabazaar_df)
plt.title('Credit Utilization Ratio Grouped By Credit Score')
plt.xlabel('Credit Utilization Ratio')
plt.ylabel('Credit Score')
plt.show

<function matplotlib.pyplot.show(close=None, block=None)>
```



1. Why did you pick the specific chart?

Clearly shows how utilization varies by group.

2. What is/are the insight(s) found from the chart?

Poor scorers have the highest utilization ratio, while good scorers keep it low.

3. Will the gained insights help creating a positive business impact?

Can be used as an early-warning KPI to flag risky customers.

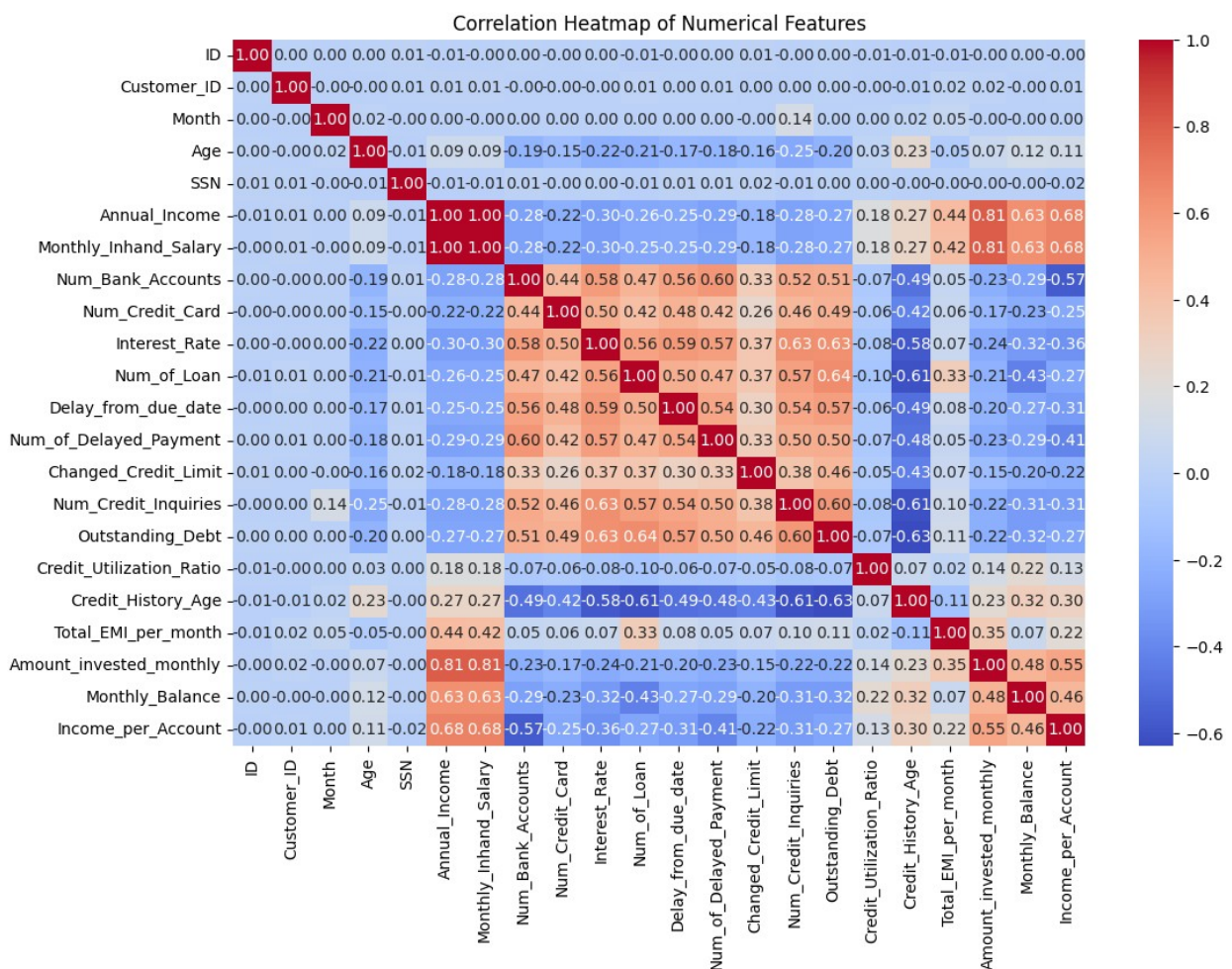
Chart - 13 Correlation Between Numerical Features - Heatmap (Risk & Fraud Indicator)

```
# Chart - 13 visualization code
# Select only numerical features
num_df = paisabazaar_df.select_dtypes(include=['int64', 'float64'])

# Compute correlation matrix
corr = num_df.corr()

# Plot heatmap
plt.figure(figsize=(12,8))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm", cbar=True)

plt.title("Correlation Heatmap of Numerical Features")
plt.show()
```



1. Why did you pick the specific chart?

Heatmaps are ideal for showing correlations between multiple numerical features.

2. What is/are the insight(s) found from the chart?

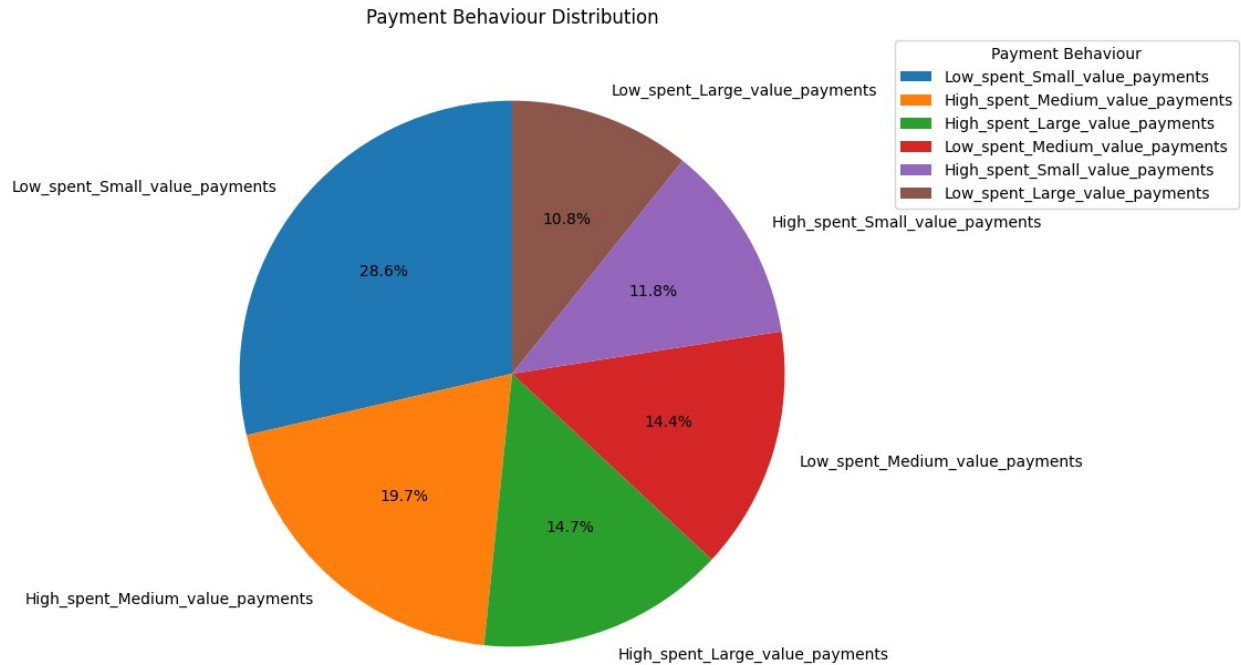
Strong positive correlation between Annual Income & Monthly Salary, and between Debt & EMI. Credit Utilization Ratio is strongly tied to credit score.

3. Will the gained insights help creating a positive business impact?

Helps in feature selection for predictive models, reducing redundancy and improving accuracy in fraud detection

Chart - 14 - Payment Behaviour Distribution - Donut Chart (Behavioural Patterns)

```
payment_behaviour_counts =  
paisabazaar_df['Payment_Behaviour'].value_counts()  
  
# Create a pie chart  
plt.figure(figsize=(8, 8))  
plt.pie(payment_behaviour_counts,  
labels=payment_behaviour_counts.index, autopct='%1.1f%%',  
startangle=90)  
plt.title('Payment Behaviour Distribution')  
  
# Add legend  
plt.legend(title='Payment Behaviour', bbox_to_anchor=(1.05, 1),  
loc='upper left')  
  
plt.show()
```

1. Why did you pick the specific chart?

Pie charts are effective for proportional breakdowns.

2. What is/are the insight(s) found from the chart?

The dataset shows a sizable share of customers with delayed or partial payments.

3. Will the gained insights help creating a positive business impact?

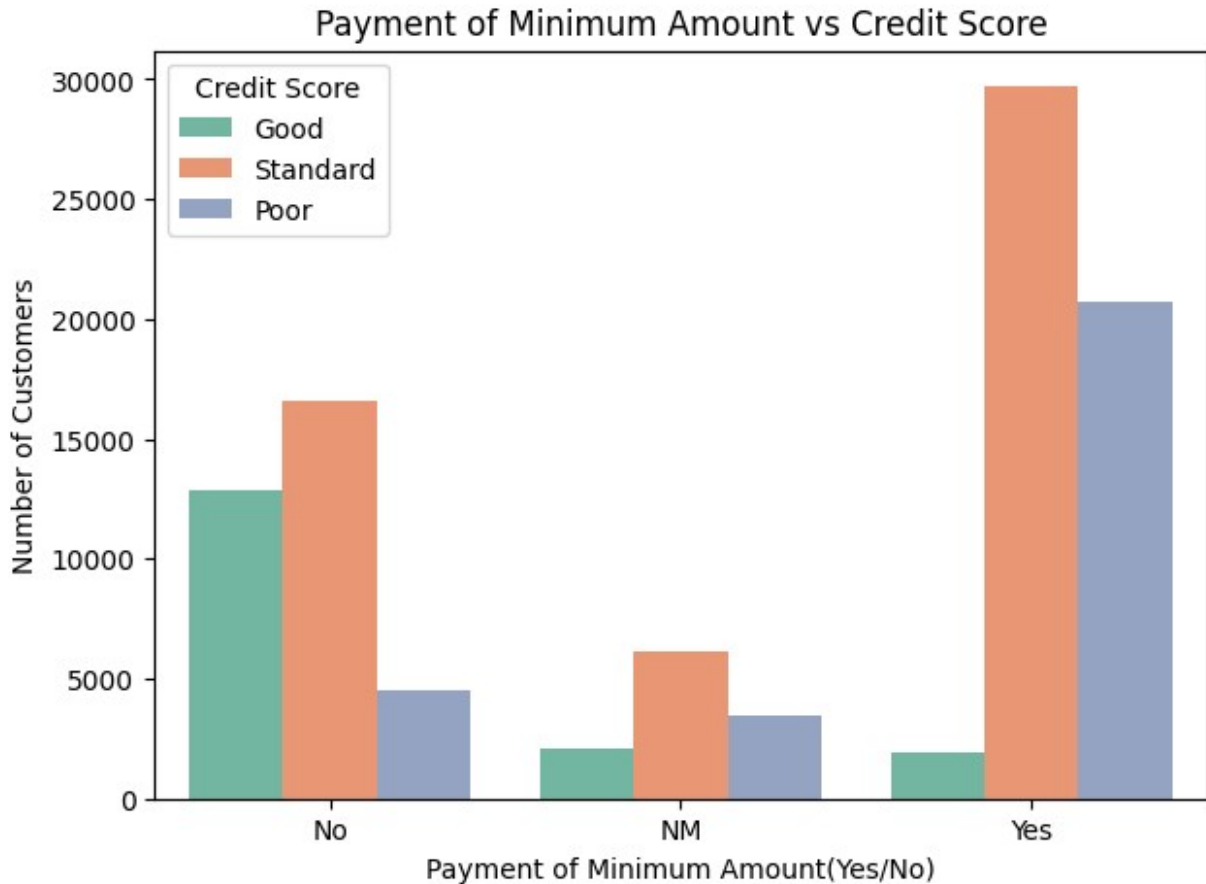
Highlights repayment culture, helping design better repayment reminder systems.

Chart - 15 - Payment of Minimum Amount(Yes/No) VS Credit Score - Bar Chart (Behavioural Patterns)

```
# Pair Plot visualization code
#Payment of Minimum Amount(Yes/No) VS Credit Score - Bar Chart
plt.figure(figsize=(7,5))

sns.countplot(data=df, x='Payment_of_Min_Amount', hue='Credit_Score',
palette='Set2')

plt.title("Payment of Minimum Amount vs Credit Score")
plt.xlabel("Payment of Minimum Amount(Yes/No)")
plt.ylabel("Number of Customers")
plt.legend(title="Credit Score")
plt.show()
```



1. Why did you pick the specific chart?

A countplot with hue (Credit Score) clearly shows how repayment habits differ across score categories.

2. What is/are the insight(s) found from the chart?

Customers who only pay the minimum amount are mostly in the Poor score group.

2. What is/are the insight(s) found from the chart?

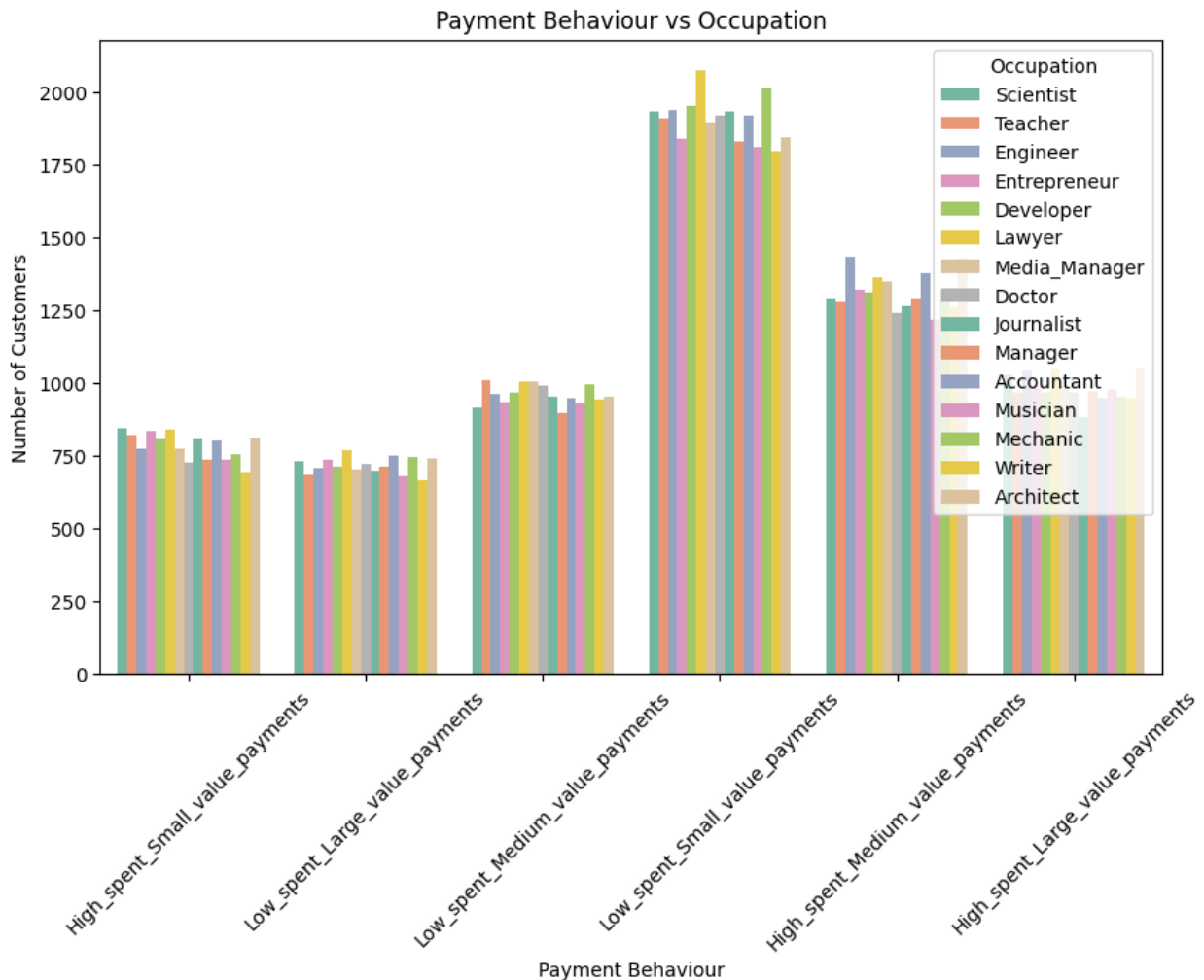
Highlights risky repayment habits, allowing financial institutions to flag these customers earlier.

Chart - 16 Payment Behaviour VS Occupation - Stacked Bar Chart (Behavioural Patterns)

```
plt.figure(figsize=(10, 6))
sns.countplot(data=paisabazaar_df, x='Payment_Behaviour',
hue='Occupation', palette='Set2')
plt.title("Payment Behaviour vs Occupation")
plt.xlabel("Payment Behaviour")
plt.ylabel("Number of Customers")
plt.legend(title="Occupation")
```



```
plt.xticks(rotation = 45)
plt.show()
```



1. Why did you pick the specific chart?

Stacked bars allow cross-comparison of repayment habits by profession.

1. What is/are the insight(s) found from the chart?

Certain occupations are more prone to delayed/partial payments.

1. Will the gained insights help creating a positive business impact?

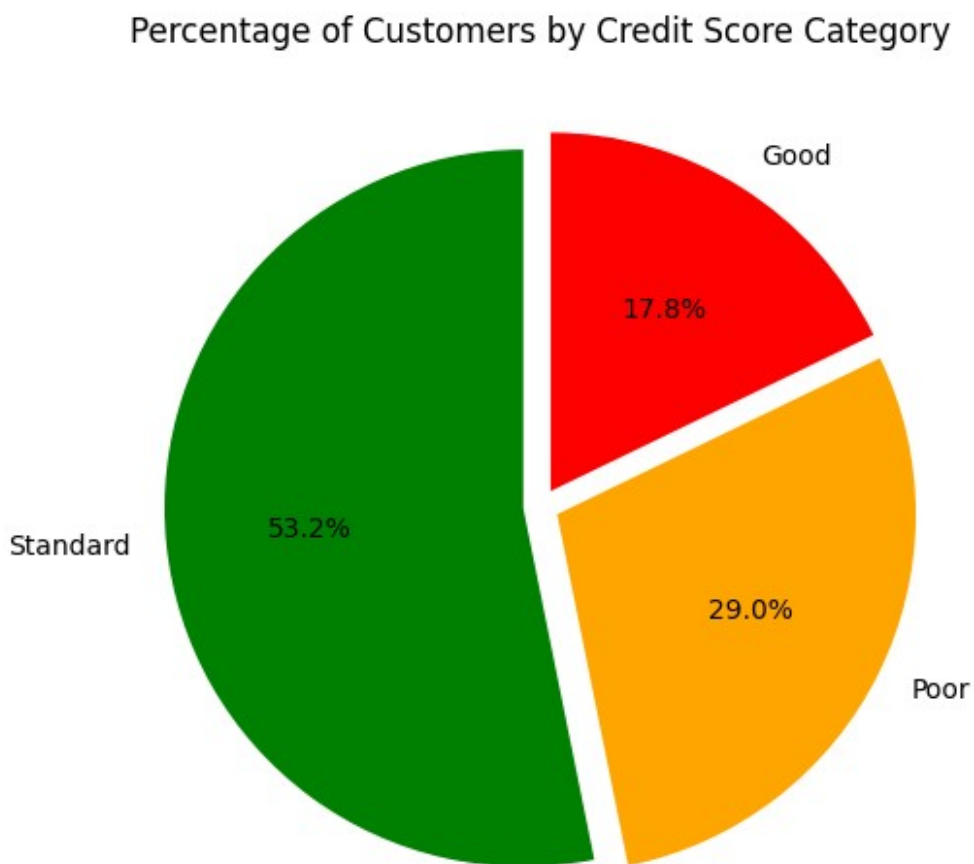
Helps in occupation-based risk profiling and customized loan eligibility rules.

Chart - 17 Percentage of Customers In Good, Standard, Poor categories - Pie Chart (Overall Credit Score Analysis)

```
# Count customers in each credit score category
credit_score_counts = paisabazaar_df['Credit_Score'].value_counts()
```

```
# Pie chart
plt.figure(figsize=(6,6))
plt.pie(credit_score_counts,
        labels=credit_score_counts.index,
        autopct='%1.1f%%',
        startangle=90,
        colors=['green', 'orange', 'red'],
        explode=(0.05, 0.05, 0.05)) # slight separation for clarity

plt.title("Percentage of Customers by Credit Score Category")
plt.show()
```



1. Why did you pick the specific chart?

A pie chart is best to show the percentage split of customers across Good, Standard, and Poor credit score categories.

1. What is/are the insight(s) found from the chart?

Majority of customers fall under the Standard category, with Poor scorers forming a significant portion, highlighting risk pockets.

1. Will the gained insights help creating a positive business impact?

Helps the bank identify the proportion of high-risk customers, so they can design stricter credit policies and focus retention efforts on good scorers.

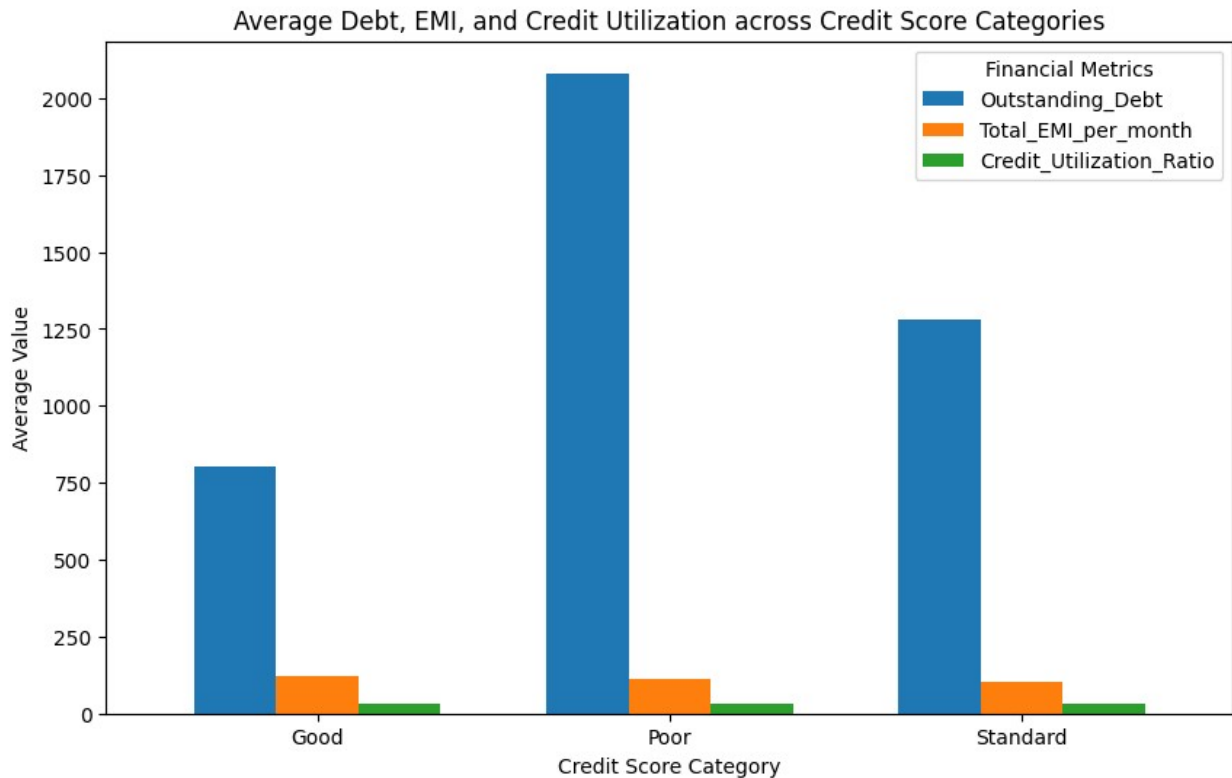
Chart - 18 Average Debt, EMI, and Utilization across Credit Score Categories - Bar Chart (Overall Credit Score Analysis)

```
# Calculate averages across Credit Score categories
avg_metrics = paisabazaar_df.groupby('Credit_Score')
[['Outstanding_Debt',
                                     'Total_EMI_per_month',
                                     'Credit_Utilization_Ratio']].mean()

# Plot grouped bar chart
avg_metrics.plot(kind='bar', figsize=(10,6), width=0.7)

plt.title("Average Debt, EMI, and Credit Utilization across Credit
Score Categories")
plt.xlabel("Credit Score Category")
plt.ylabel("Average Value")
plt.legend(title="Financial Metrics")
plt.xticks(rotation=0)

plt.show()
```



1. Why did you pick the specific chart?

Grouped bar charts allow simultaneous comparison of multiple financial KPIs across categories.

1. What is/are the insight(s) found from the chart?

Poor scorers consistently have higher EMI and utilization ratios compared to Good scorers.

1. Will the gained insights help creating a positive business impact?

Confirms which financial metrics drive poor credit scores, enabling early warning models and better fraud detection

6. Solution to Business Objective

What do you suggest the client to achieve Business Objective ?

To achieve the business objective of minimizing fraud risk and enhancing the quality of the credit portfolio, a strategic and data-driven approach should be adopted based on the insights obtained from the analysis. Customers can first be categorized into different risk segments by evaluating important financial indicators such as credit score, outstanding debt, EMI burden, repayment history, and credit utilization ratio. High-risk customers exhibiting excessive debt levels, frequent delayed payments, high utilization ratios, or unstable repayment behavior should be continuously monitored through advanced risk management systems.

Financial institutions should implement proactive customer engagement strategies to encourage responsible repayment behavior. This may include automated payment reminders,

customized repayment schedules, financial awareness programs, and flexible EMI restructuring options for customers experiencing financial difficulties. Such initiatives can help reduce defaults and improve overall repayment consistency.

In addition, predictive analytics and machine learning models can be developed to identify early warning signs of fraudulent activities and potential loan defaults. Features such as repeated credit inquiries, sudden increases in credit utilization, frequent changes in credit limits, and irregular payment behavior can serve as critical indicators in fraud detection systems. Automated alert mechanisms can notify financial institutions whenever suspicious activities or high-risk behaviors are detected.

The organization should also strengthen its loan approval and credit evaluation policies by applying stricter eligibility criteria for customers classified under high-risk categories. This can help reduce exposure to bad debts and improve the overall quality of the lending portfolio. At the same time, customers with strong repayment histories and stable financial behavior should be rewarded with benefits such as lower interest rates, higher credit limits, loyalty rewards, and personalized financial products. This approach not only improves customer satisfaction but also encourages long-term customer retention.

By integrating these strategies with continuous monitoring and analytical decision-making, financial institutions can significantly reduce fraud risk, improve credit risk management, and optimize lending operations. Ultimately, these measures will contribute to higher profitability, stronger customer trust, and a more secure and sustainable financial ecosystem.

Conclusion

This project provided a comprehensive analysis of customer financial behavior, credit performance, and risk patterns using the dataset provided by Paisabazaar. Through effective data preprocessing techniques such as datatype correction, handling missing values, removing inconsistencies, and preparing the dataset for analysis, the project established a strong foundation for meaningful Exploratory Data Analysis (EDA) and visualization.

The analysis revealed several important insights related to customer creditworthiness and financial stability. Examination of the Credit Score categories showed that the majority of customers belonged to the Standard category, while customers classified under Good and Poor credit scores demonstrated significantly different financial behaviors. Customers with Good credit scores generally maintained lower outstanding debts, controlled credit utilization ratios, and timely repayment habits. In contrast, customers with Poor credit scores were associated with higher debt burdens, delayed payments, larger EMI obligations, and unstable repayment patterns.

The study of customer income groups highlighted that income alone is not sufficient to determine financial reliability. Instead, repayment discipline, debt management, and credit utilization played a much more significant role in influencing credit scores. Customers who consistently paid only the minimum due amount were observed to have a higher probability of falling into the Poor credit category. Similarly, increased delays in payments after due dates strongly indicated financial stress and elevated credit risk.

Loan and borrowing behavior analysis showed that many customers heavily relied on personal loans and credit cards. Customers managing multiple loan accounts simultaneously often exhibited weaker credit profiles and greater repayment difficulties. Correlation analysis further demonstrated strong relationships between important financial variables such as outstanding debt, annual income, EMI obligations, and credit utilization ratio. These features emerged as critical indicators for evaluating customer risk and financial health.

Visualization techniques such as bar charts, heatmaps, scatter plots, boxplots, and radar charts played an essential role in identifying hidden trends and behavioral patterns. The radar chart comparisons clearly illustrated that customers with Poor credit scores consistently performed poorly across multiple financial indicators, whereas Good credit score customers maintained balanced financial management and healthier credit behavior.

The project successfully achieved its objective of identifying major factors contributing to fraud risk and poor credit performance. Findings from the analysis suggest that behaviors such as excessive credit utilization, frequent delayed payments, high debt obligations, repeated credit inquiries, and minimum-only payments are strong warning signs of financial instability and potential fraud risk.

Overall, this project demonstrates the importance of data-driven analysis in modern banking and financial systems. The insights generated through EDA and KPI analysis can help Paisabazaar strengthen its credit risk assessment models, improve fraud detection mechanisms, optimize lending decisions, and encourage responsible borrowing behavior among customers. By leveraging these analytical insights, financial institutions can build safer, more transparent, and more efficient financial ecosystems while improving both customer trust and long-term institutional stability.